

SALES PERFORMANCE DASHBOARD

Algorithm and Dataset

1. Algorithm

Step-by-Step Procedure

1. Collect the sales dataset containing information such as Order ID, Order Date, Product, Category, Region, Sales, Quantity, Discount and Profit.
2. Import the dataset into Power BI.
3. Clean and preprocess the data by removing duplicates, handling missing values and correcting data types.
4. Transform the data using Power Query for analysis and reporting.
5. Create relationships between tables wherever required.
6. Create calculated measures and KPIs using DAX.
7. Calculate important business metrics such as Total Sales, Total Profit, Total Quantity and Profit Margin.
8. Analyze sales performance based on time, product, category and region.
9. Create interactive visualizations such as KPI cards, bar charts, line charts, tables and slicers.
10. Build the final interactive dashboard to identify sales trends and support business decision-making.

Logic / Method Used

- Data cleaning and transformation
- Aggregation and KPI calculation
- Time-based sales analysis
- Category-wise and region-wise comparison
- Filtering and interactive analysis
- Descriptive analytics for identifying sales trends

Model / Technique Applied

- **Power Query:** Used for data cleaning and transformation.
- **DAX:** Used to create calculated measures and KPIs.
- **Data Visualization:** Used to represent sales performance through charts, cards, tables and slicers.
- **Descriptive Analytics:** Used to understand historical sales performance and identify trends.

Process Flow

Raw Sales Data → Data Cleaning → Data Transformation → Data Modeling → DAX Calculations → KPI Analysis → Visualization → Interactive Sales Dashboard

2. Dataset

Dataset Attribute	Details
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Source of Data	Publicly available sales dataset (Sample Superstore / sales dataset)
Number of Records / Samples	9,994 sales records
Data Format	CSV / Excel
Training and Testing Split	Not Applicable – this is a Business Intelligence and descriptive analytics project.

Features / Attributes (Columns)

- Order ID
- Order Date
- Customer Name / ID
- Product Name
- Category
- Sub-Category
- Region
- Sales
- Quantity
- Discount
- Profit

Data Types

- **Numerical:** Sales, Quantity, Discount, Profit
- **Categorical:** Product, Category, Sub-Category, Region, Customer
- **Date:** Order Date
- **Identifier:** Order ID / Customer ID

Conclusion: The processed sales dataset is used to calculate business KPIs and generate interactive visualizations that help users understand sales trends, product performance, regional performance and profitability.